

2026 PNU TECH WEEK · PHYSICAL AI HACKATHON

Autonomous Search & Rescue

A TurtleBot3 that maps an unknown apartment, finds two red apples and drives back to where it started, all on its own.

Team Alien

챗제이야 · 찻수뽀이 · 텃까웅산 · 손엘래나



WHO WE ARE

Team Alien



첵제이야



치수뛰이



텃까웅산



손엘래나

Built with

Webots R2025a

TurtleBot3 Burger

Python

OpenCV

NumPy

Ultralytics YOLO11n

THE CHALLENGE

Rescue two red apples

- 1 Find 2 red apples on the floor
- 2 Confirm each one up close
- 3 Drive back to the start point

WHAT MAKES IT HARD

- No map: only the start pose is known
- Decoys: green, purple and orange apples, a red can
- A walking pedestrian, furniture, carpet edges



The Webots apartment world the robot has to search

OUR CONTRIBUTION

What we were given, what we built

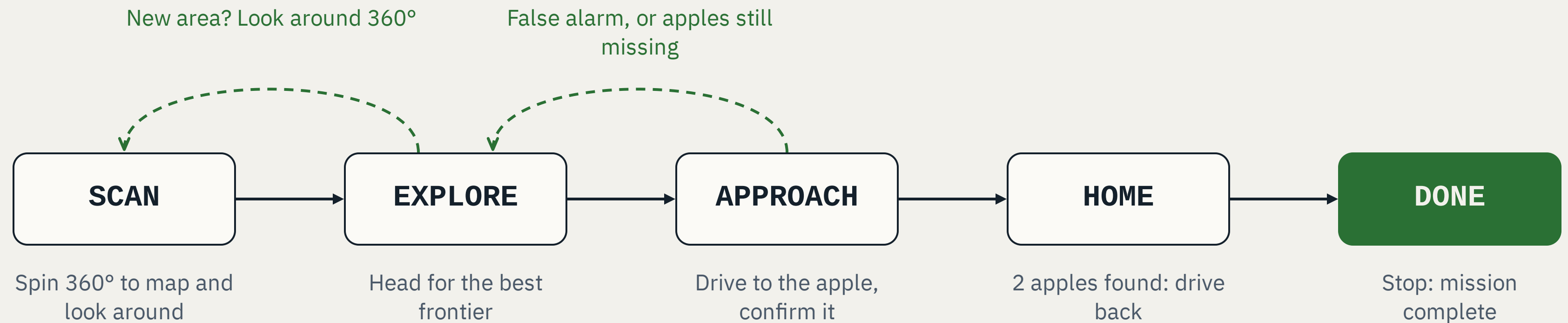
GIVEN • STARTER KIT

Keyboard driving demo
Sensor and camera demos
Color segmentation demo
YOLO11n model + demo
Ground-truth pose viewer
Apartment world + apples
Breakroom test worlds

BUILT BY US • `tb3_sar` • ~1,200 lines

- ✓ Mission state machine
- ✓ Compass calibration + scan matching
- ✓ Occupancy grid + camera coverage map
- ✓ Semantic object map + frontier scoring
- ✓ Red-apple detector with 6 checks
- ✓ A* + pure pursuit + DWA planner
- ✓ Safety stop, tilt sensing, corner escape
- ✓ Live map, camera and costmap views

Five states, one mission



Safety layer, every step: stop under 20 cm, back off, feel low obstacles by tilt.

HOW IT WORKS

Inside the controller

SENSE

Wheel encoders

Compass

Accelerometer

360° LiDAR

Camera

THINK

Localize

odometry + compass +
LiDAR scan matching

Perceive

red-apple detector + YOLO
object map

Map

log-odds occupancy grid, 5
cm cells

Plan

frontier choice + A* path

ACT

Pure pursuit

DWA planner

Wheel speeds

Safety + recovery

Knowing where we are

01

Odometry + compass

Wheels measure distance; the compass, calibrated on the first spin, gives heading.

02

LiDAR scan matching

Nudge the pose up to ± 15 cm until the scan lines up with known walls.

03

Occupancy grid

A 40×40 m map around the start, 5 cm cells, built from every scan.

4.49 m

of position drift corrected by 247 scan-match fixes in one 8-minute test run.

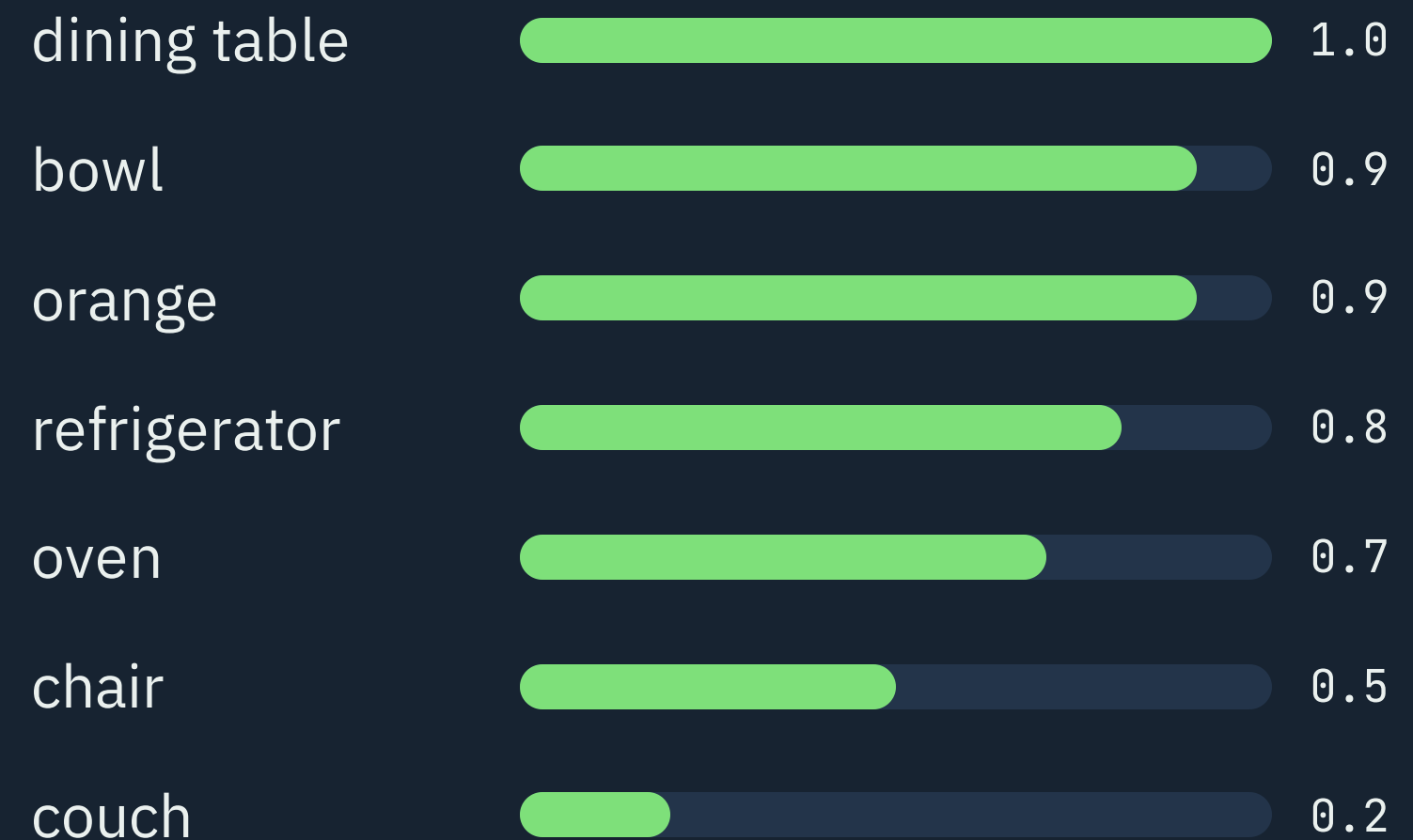
Where would an apple be?

- 01 Frontiers are map edges and floor the camera hasn't checked yet.
- 02 YOLO names objects; the LiDAR range pins them on the map.
- 03 Pick the frontier that is close and near kitchen-like things.

$\text{score} = \text{path} - 3.0 \times \sum \text{prior} \times \text{share} \times e^{(-d / 1.5 \text{ m})}$

Lowest score wins. In a test run it picked $(-0.3, -9.6)$ over the nearest frontier $(-1.1, -10.6)$.

Apple prior per object



Is it really a red apple?

01 **Red** — inside the red hue band

02 **Round** — circular, close to square

03 **On the floor** — below the horizon line

04 **Right size** — fits its distance on the floor

05 **Whole** — not cut off at the image edge

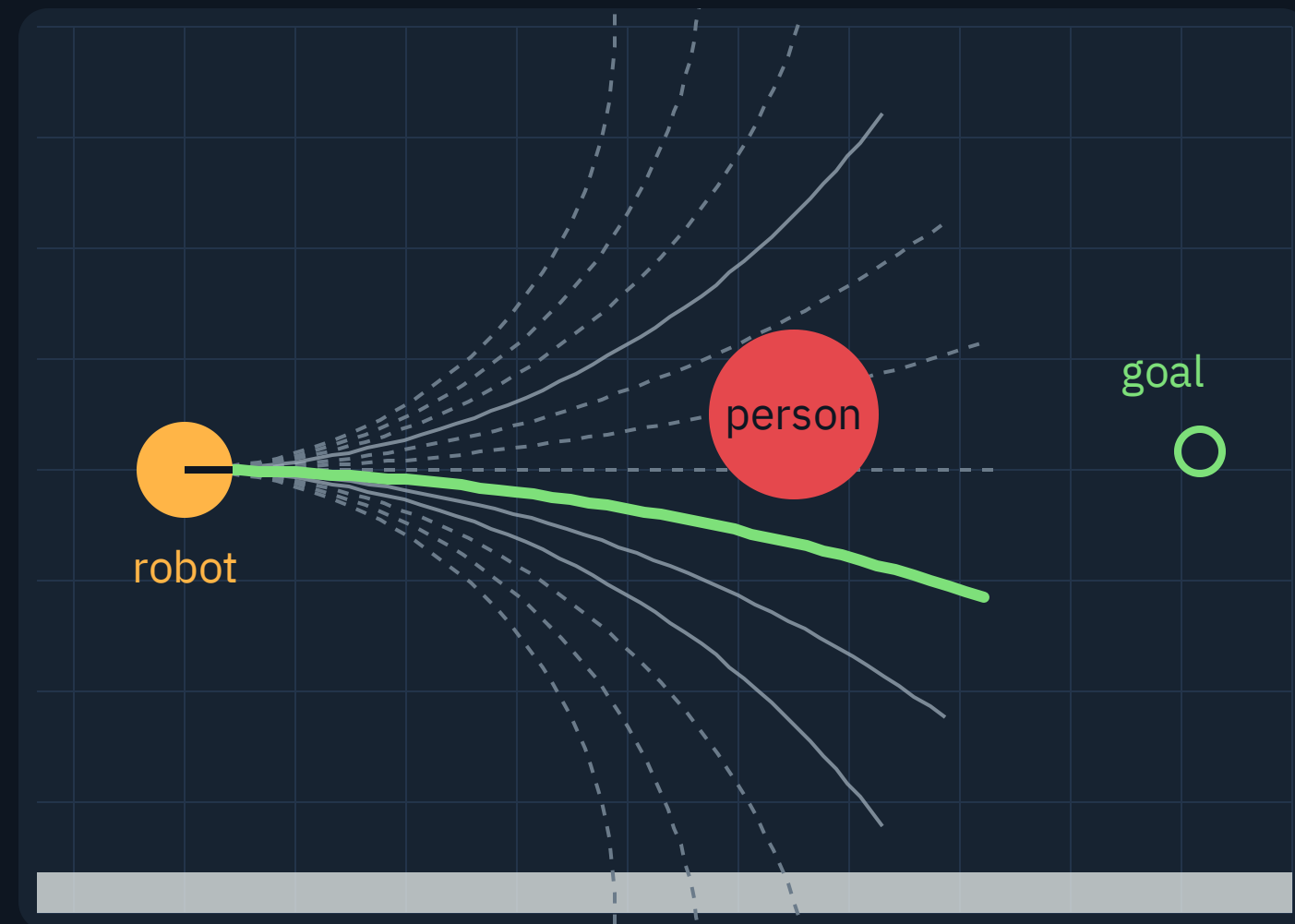
06 **Not tall** — no red continuing above it

3 matching sightings before the robot switches to APPROACH

2 close-up confirmations under 2 m before an apple counts

0 false alarms in our test runs, despite four decoys

Driving safely around people



Grey: safe options · dashed: would collide · green: chosen

GLOBAL · A*

A* on a map where walls are padded by the robot's size; re-planned every 1.5 s.

LOCAL · DWA

Tests 90 trajectories 1.5 s ahead on a live 3 m costmap, so people leave no ghosts.

SAFETY · every step

Under 20 cm: stop, wait, back off, re-plan. No progress in 3.5 s: recover.

Seeing what the LiDAR can't

15 cm height of the LiDAR scan
above the floor

5 mm height of a carpet edge:
invisible to it

01

See it

The camera spots small coloured objects on the floor and marks them before contact.

02

Feel it

A tilt over 6° for 0.3 s means a carpet edge the camera can't see.

03

Back off & remember

Reverse until level, then mark the spot so A* and DWA avoid it.

04

Escape

3 back-offs in 15 s: turn to the most open direction, pick a new goal.

What testing taught us

2 / 2

red apples found and confirmed

0

false alarms, despite four decoys

5

problems found in testing and fixed

Problem in testing

- Fell over on a carpet edge
- Missed the apple under a sink
- Kept backing into a corner
- Counted a found apple twice
- Kept reversing next to home

Fix

- Tilt sensing with the accelerometer
- Failed spots are retried, not ignored for good
- Back-off limit, then turn to open space
- Close-range positions, rescued apples remembered
- Home reached within 0.6 m when a wall blocks

VISION

Next: a guide robot for blind people

OUR ROBOT TODAY

Explores with no prior map

Dodges walking people live

Sees and feels low obstacles

Finds objects with its camera

Remembers the way back

A GUIDE ROBOT

→ **Guides through unfamiliar buildings**

→ **Walks safely through crowds**

→ **Warns about steps, curbs, trip hazards**

→ **Finds doors, seats, lost keys**

→ **Leads its user home**

STILL TO ADD

Voice + haptic handle

Depth camera for stairs

Walking pace, person-sized footprint

Outdoor: GPS, crosswalks

TEAM ALIEN

Thank you

Questions?

챗제이야 · 찻수뽕이 · 텃까웅산 · 손엘래나

github.com/Xuf313/hackathon